Quant

Group 6

Rating correlations

GPT4	r	RMSE
Criteria × Questions	0.393	0.987
Criteria	0.835	0.451
Questions	0.22	0.71

Table 1: Median Pearson's r and root mean square error values for GPT4 ratings

GPT5	r	RMSE
Criteria × Questions	0.15	1.24
Criteria	0.554	0.667
Questions	−0.007	0.805

Table 2: Median Pearson's r and root mean suqare error values for GPT 5 ratings

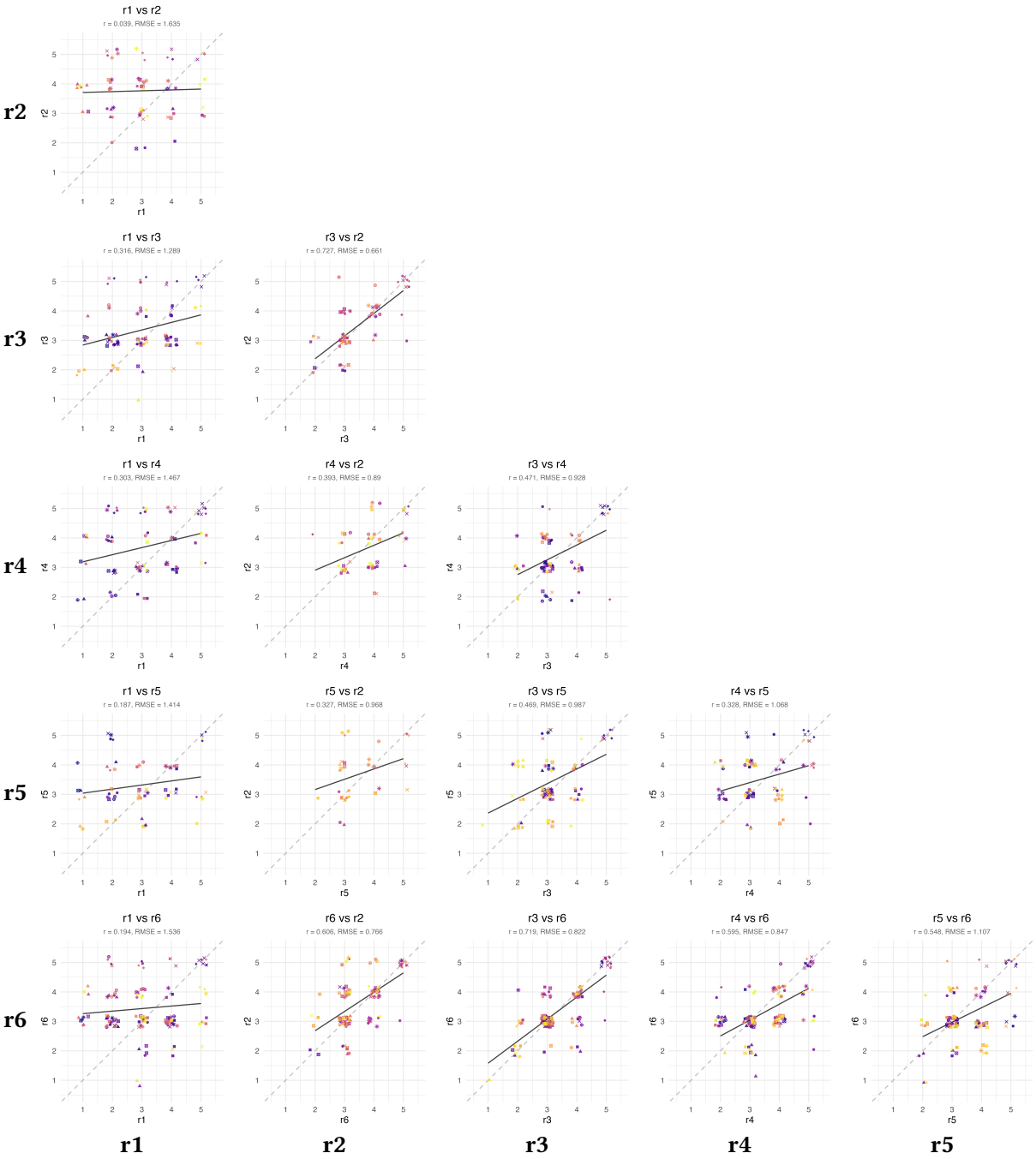


Figure 1: Pairwise agreement plots: Q×C level, GPT4

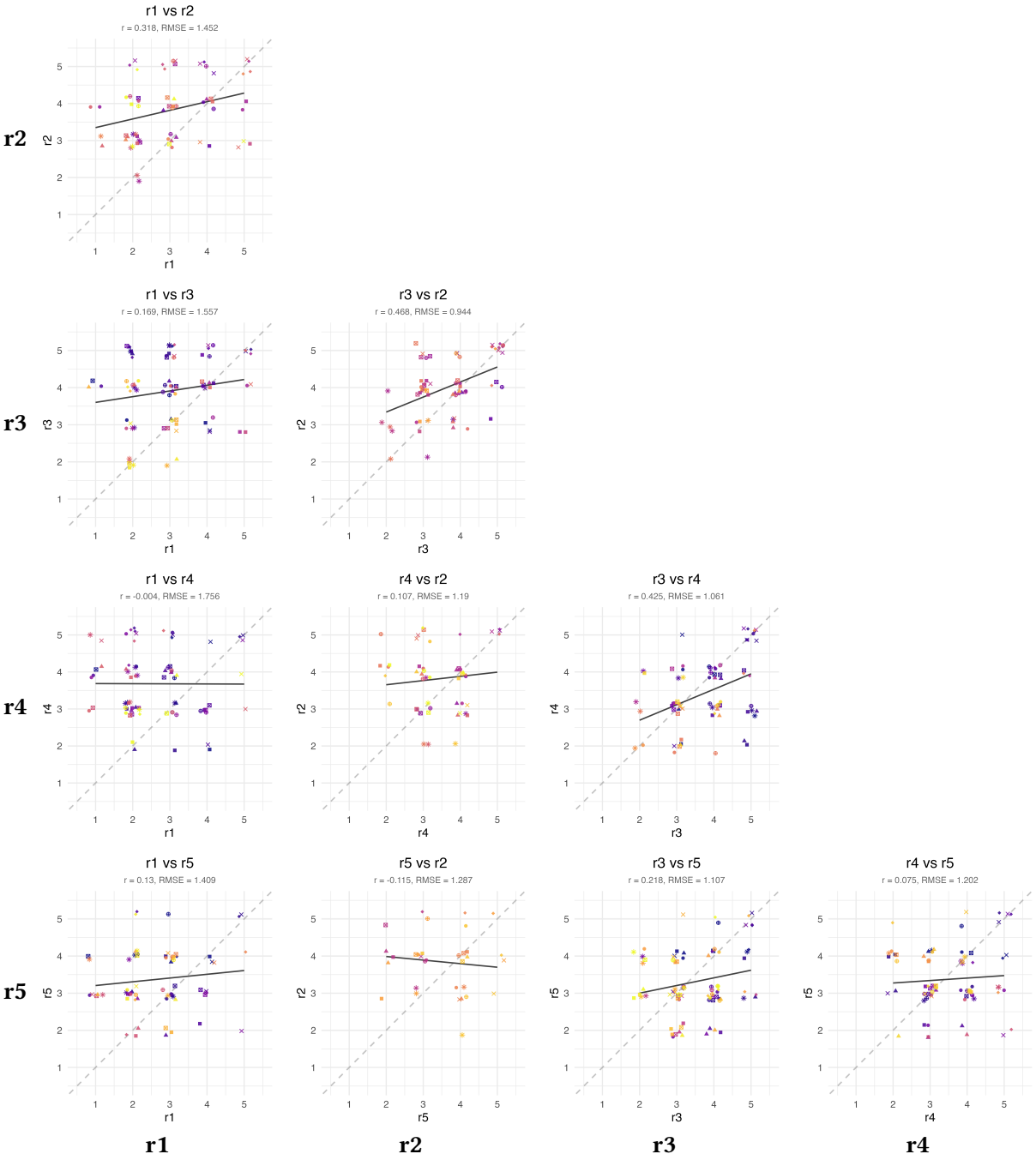


Figure 2: Pairwise agreement plots: Q×C level, GPT5

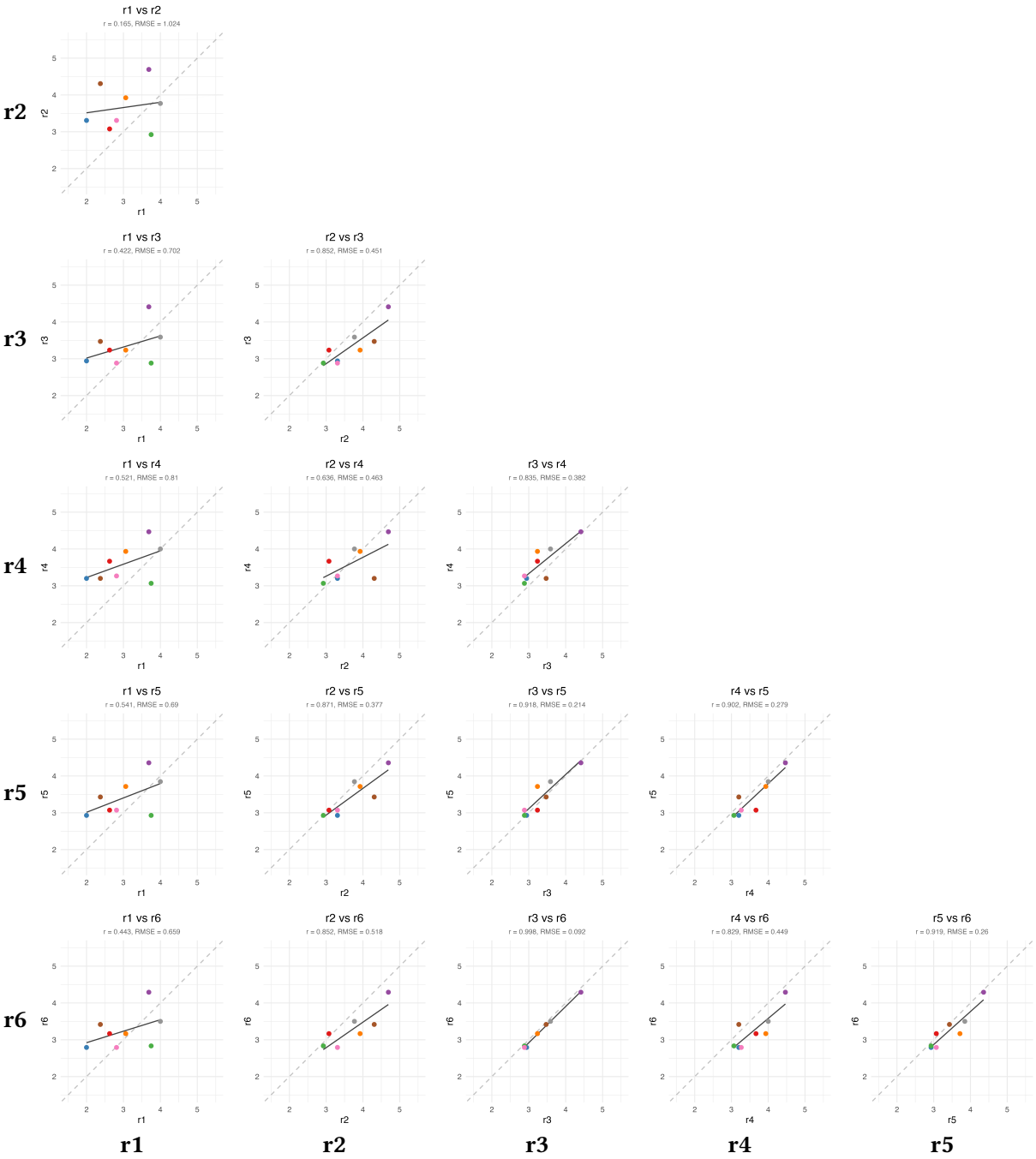


Figure 3: Pairwise agreement plots: criteria level, GPT4

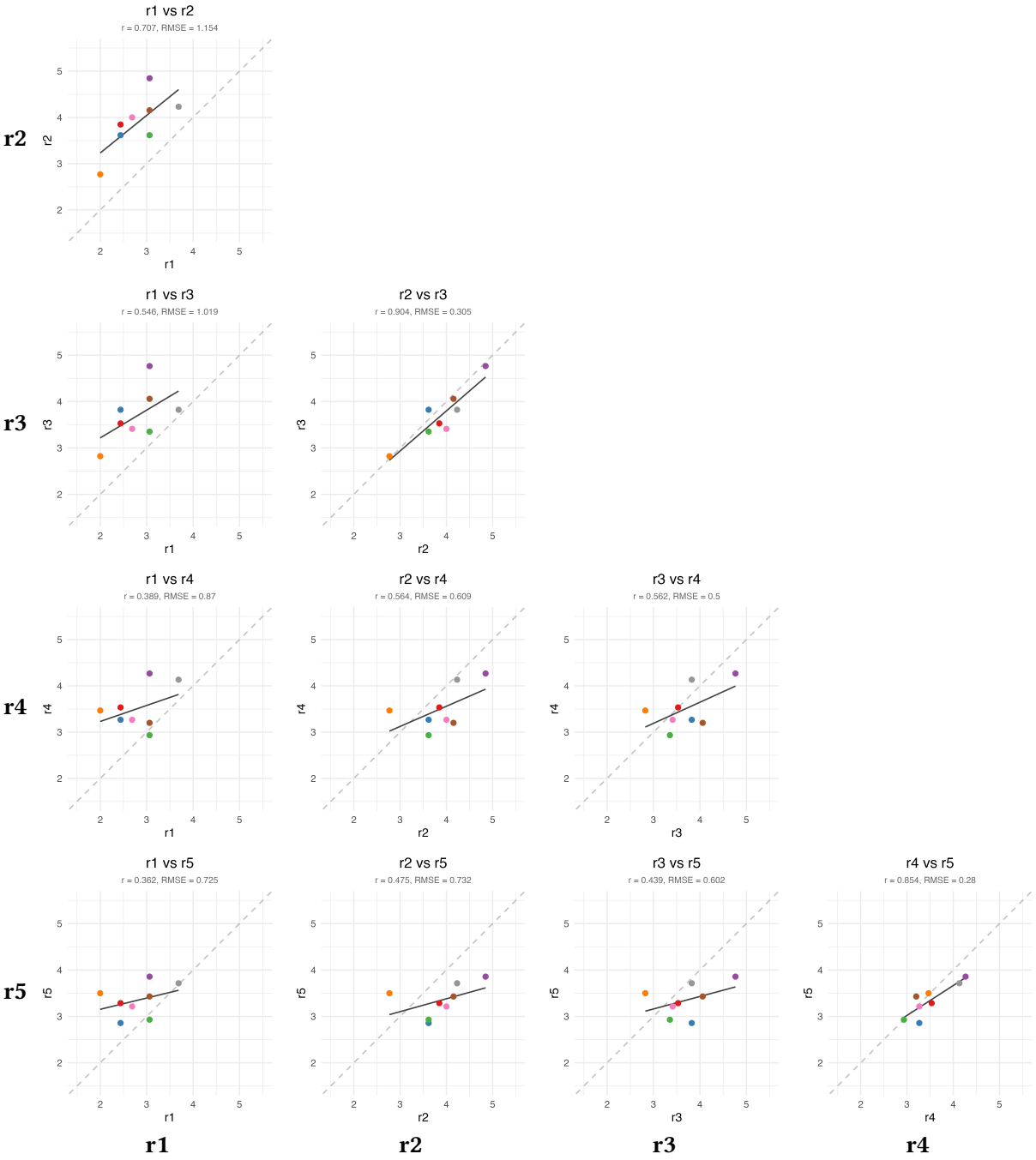


Figure 4: Pairwise agreement plots: criteria level, GPT5

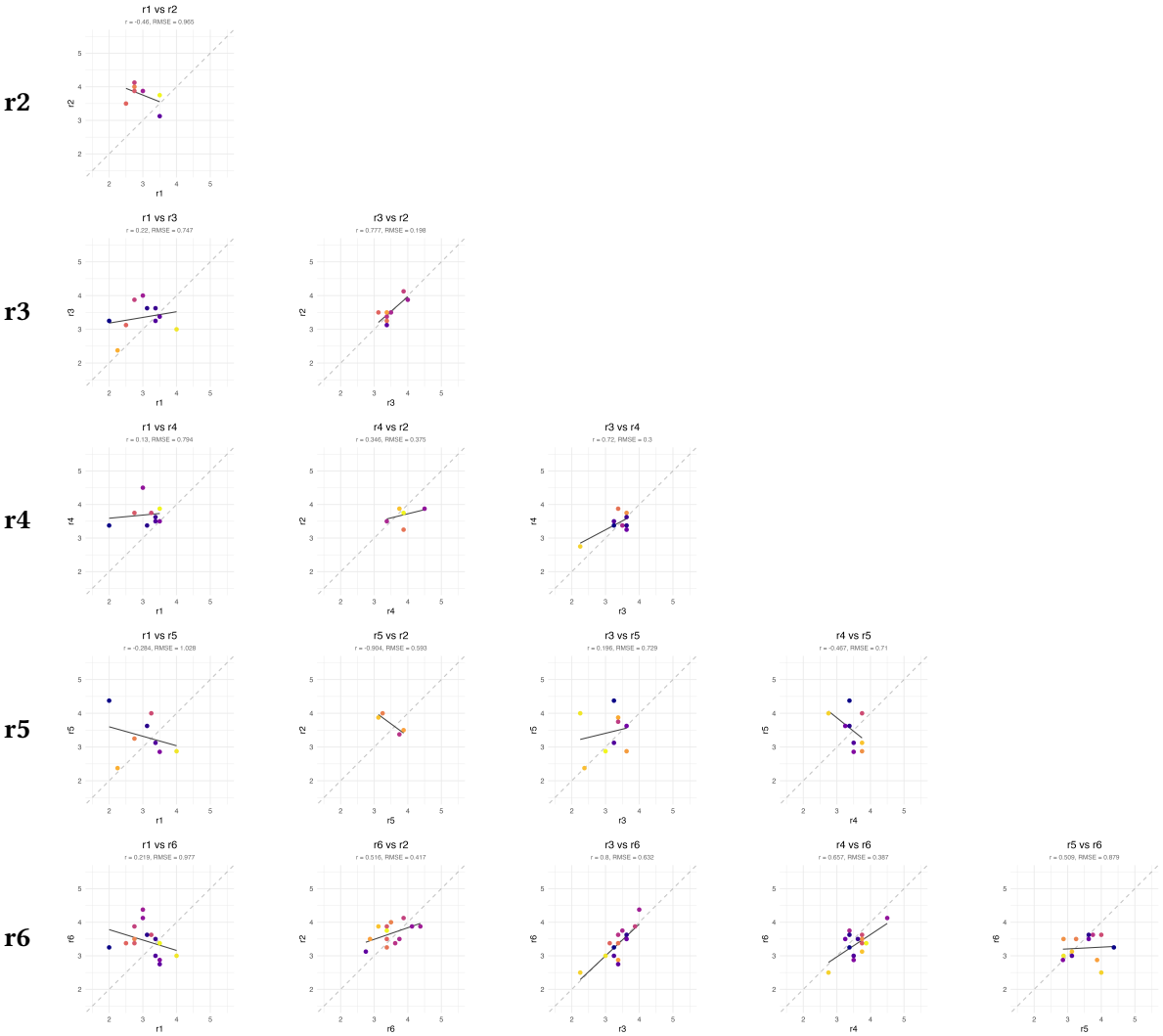


Figure 5: Pairwise agreement plots: question level, GPT4

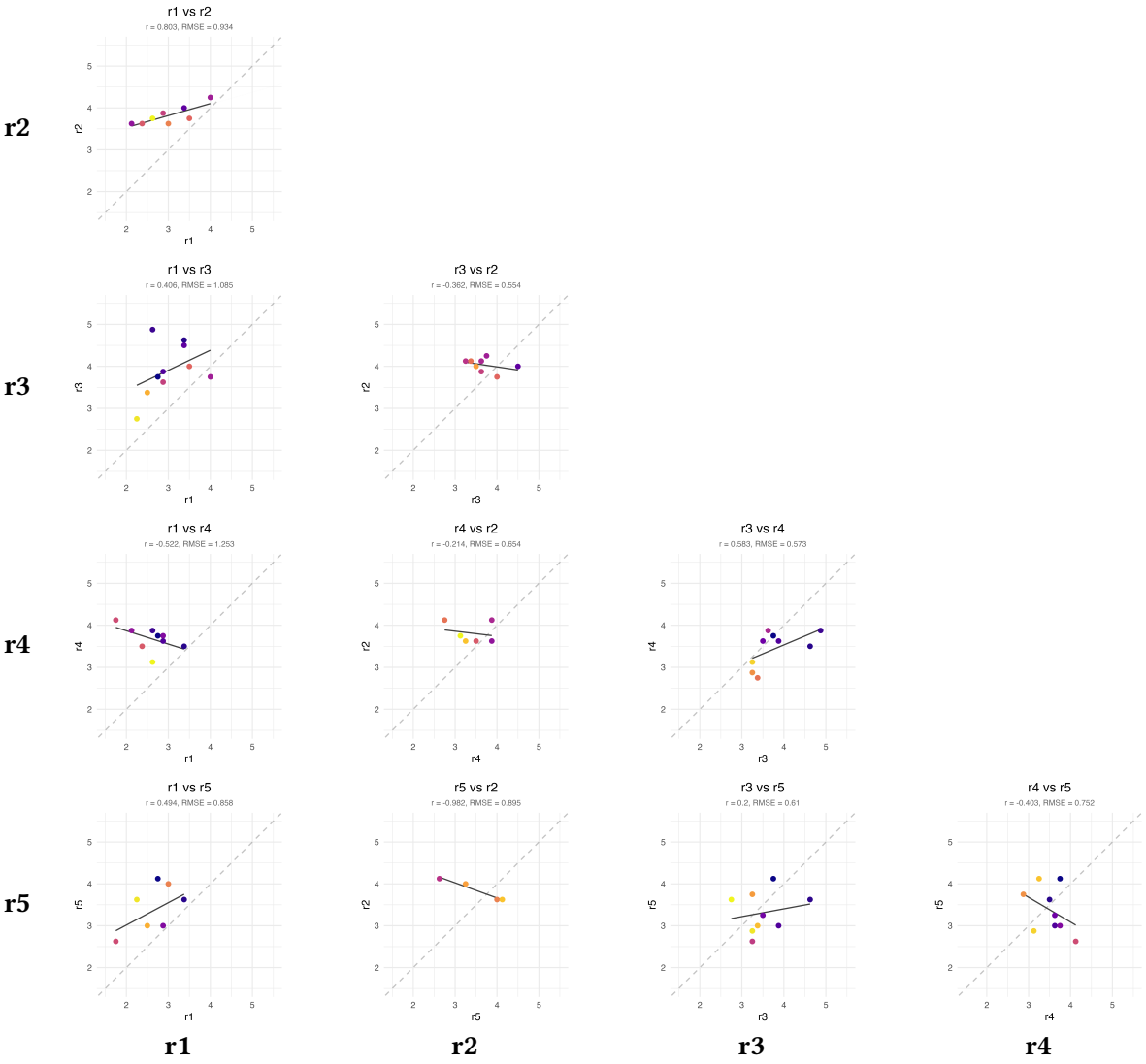


Figure 6: Pairwise agreement plots: question level, GPT5

Paired samples t-test

The Irrelevance criterion was excluded from inferential analyses due to a skewed distribution (ceiling effect), as inspection of its histogram revealed ratings concentrated at the higher end of the scale, violating the normality assumption.

A paired samples t-test was conducted to compare ratings between GPT-4 (M = 3.31, SD = 0.949) and GPT-5 (M = 3.35, SD = 0.955) across seven criteria. No significant overall difference was found ($t(565) = -1.04$, p = .299, mean diff = −0.049, 95% CI [−0.143, 0.044]).

Per-criterion analyses were conducted with Bonferroni correction applied(α = .05/7 = .007). Two criteria showed significant differences after correction: GPT-4 was rated significantly higher on Narrativity ($t(74) = 5.22$, $p < .001$, $d = 0.603$), while GPT-5 was rated significantly higher on Factuality ($t(74) = -3.20$, p = .014, d = −0.370). Both represent medium effect sizes. Recall and Reduction showed uncorrected significant differences that did not survive Bonferroni correction (both $p > .30$ after correction).

A limitation of this study is the small number of raters (n = 5). However, given the Central Limit Theorem, the aggregated ratings are expected to approximate a normal distribution, supporting the validity of parametric testing.

Discard: Inter-Rater Reliability

Inter-rater reliability was assessed using Krippendorff's α (ordinal) across six raters for GPT-4, five raters for GPT-5, and eight evaluation criteria. Overall agreement was below the commonly accepted threshold of 0.667 [@krippendorff] for both models (GPT-4 α = 0.372, GPT-5 α = 0.117; pg: 11). Agreement varied considerably across criteria: Ambiguity and Narrativity showed near-zero or negative α values for both models, indicating systematic disagreement, while Repetition showed the highest agreement for GPT-4 (α = 0.543), approaching but not reaching the acceptable threshold. GPT-5 agreement remained low across all criteria (pg: 11–12). A sensitivity analysis excluding each rater individually showed that no single rater was responsible for the low agreement — the most impactful exclusion (R3) improved GPT-4 α from 0.372 to 0.535 and GPT-5 α from 0.117 to 0.216 (pg: 12–15). This suggests the disagreement is structural, likely reflecting the inherent subjectivity of certain criteria rather than individual rater error. Given this, all five raters were retained for subsequent analyses.